

LivoSpeak AI — System Architecture & DPDP Compliance

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This document details the production-ready system architecture, pronunciation scoring algorithms, and security guidelines for LivoSpeak AI. It details our architectural trade-offs, model selection, and comprehensive compliance strategy with India's Digital Personal Data Protection (DPDP) Act 2023.

1. System Components & Connections

LivoSpeak AI uses an asynchronous, decoupled client-server architecture designed for high-performance, low-latency analysis, and complete data privacy.

[PLACEHOLDER: Paste System Architecture Diagram (mermaid-drawing.svg) Here]

- Frontend (Next.js 15): Handles recording wrapper & dynamic speaking statistics dashboard.
- Backend (FastAPI): API gateway, orchestrates audio transcoding, Groq API calls, and scoring.
- Database (MongoDB): Persists speaking transcripts, analytics history, and practice plans.
- Transcoding (FFmpeg): Standardizes browser WebM/OGG files to linear 16kHz mono WAV.

2. Model Selection & Rationale

To scale the platform on lightweight hosting without GPU dependency, LivoSpeak offloads inference workloads to Groq's high-speed LPU platform:

Capability	Model Selected	Core Rationale over Alternatives
Speech-to-Text	Groq Whisper Large V3	Sub-second latency; delivers precise word-level confidence and timestamp alignments.
Speech Coaching	Llama 3.3 70B	Superb instruction-following in JSON mode; complies with zero user data retention.
Database Driver	Motor (Async)	Keeps the FastAPI event loop fully non-blocking during DB read/write history operations.

3. Pronunciation Scoring & Highlight Logic

Speech-to-text models automatically correct accents. To provide accurate feedback, LivoSpeak uses a Hybrid STT-LLM Scoring Pipeline:

- **Pronunciation Score:** Calculates base from Whisper word confidence scores, then subtracts 6 points for each real pronunciation mistake identified by Llama 3.3 (up to 30 points max).
- **Speech Clarity:** Measures high-confidence token ratios, applying 5-point deductions per articulation error.
- **Fluency:** Evaluates speaking rate against the native English range (110-145 WPM). Points are deducted for speed deviations and silent pauses over 1.2s (-4 points per pause).
- **Visual Highlights:** Identified mistake words are compared to the token array using punctuation-stripped string matching and rendered in red on the interactive UI transcript.

4. India DPDP Act 2023 Compliance

LivoSpeak AI enforces privacy-by-design to align with India's DPDP Act 2023 guidelines:

- **Zero Audio Retention:** Audio files are processed in-memory or written to secure temp folders, then physically deleted (`os.remove()`) inside a python finally block immediately after analysis. Audio data is kept for < 5 seconds.
- **Biometric Safeguards:** The platform never generates or stores voiceprints, biometric parameters, or voice recordings.
- **Consent:** Recording and upload triggers serve as clear affirmative consent.
- **Data Residency:** All backend endpoints and MongoDB clusters run inside compliant regional cloud data centers.
- **Right to Erasure:** Users can delete specific speaking sessions via the dashboard trash icon, triggering an immediate DB record purge.

5. Architectural Trade-offs & Next Steps

- **Trade-off: Ephemeral Audio:** Because audio is deleted instantly for security, past recording audio cannot be replayed. This eliminates the risk of voice biometric profile exposure.
- **Next Steps:** Integrate OAuth2 user auth (NextAuth/Auth0), add real-time acoustic phoneme mapping (e.g. Wav2Vec2), and add comparative progress trend graphs.